
Dark portal
$L_e - L_\alpha$
Secret interactions, vector
Secret interactions, scalar, Majorana
Secret interactions, scalar, Dirac
ν_α
$\bar{\nu}_\beta$
Z_D
$g_D\, U_{\alpha 4} U^*_{\beta 4}$
ℓ, ν_ℓ
$\bar{\ell}$, $\bar{\nu}_\ell$
Z'
$+g'$ $(\ell = e)$, $-g'$ $(\ell = \alpha)$
V
$G_{\alpha\beta}$
ϕ
$G_{\alpha\beta}$
$\nu_{L\alpha}$
$\bar{\nu}_{R\beta}$
$G_{\alpha\beta}$, $\nu_L \to \nu_R$